Software Requirements Specification for Charged Particle Trajectory Simulator

Zhuo Zhang

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Contents

1 Reference Material

This section records information for easy reference.

1.1 Table of Units

The unit system used throughout is SI (Système International d'Unités). In addition to the basic units, several derived units are also used. For each unit, the Table of Units lists the symbol, a description, and the SI name.

Table 1: Table of Units

Symbol	**Description**	**SI Name**
C	electric charge	coulomb
kg	mass	kilogram
m	length	metre
N	force	newton
s	time	second
T	magnetic flux density	tesla

1.2 Table of Symbols

The symbols used in this document are summarized in the Table of Symbols along with their units. Throughout the document, symbols in bold will represent vectors, and scalars otherwise. The symbols are listed in alphabetical order. For vector quantities, the units shown are for each component of the vector.

Table 2: Table of Symbols

Symbol	**Description**	**Units**
a_{x}	X-acceleration of the particle	$\frac{\text{m}}{\text{s}^2}$
a_{y}	Y-acceleration of the particle	$\frac{\text{m}}{\text{s}^2}$
$\mathbf{a}(t)$	Acceleration	$\frac{\text{m}}{\text{s}^2}$
B	Out-of-plane magnetic flux density	T
B_{max}	Maximum magnetic flux density	T
E_{max}	Maximum electric field magnitude	$\frac{\text{N}}{\text{C}}$
E_{x}	X-component of the electric field	$\frac{\text{N}}{\text{C}}$
E_{y}	Y-component of the electric field	$\frac{\text{N}}{\text{C}}$

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Table 2: Table of Symbols (Continued)

Symbol	**Description**	**Units**
$\mathbf{F}$	Force	N
m	Particle mass	kg
m_{max}	Maximum particle mass	kg
m_{min}	Minimum particle mass	kg
$\mathbf{p}(t)$	Position	m
q	Particle charge	C
q_{max}	Maximum charge magnitude	C
t	Time	s
t_{final}	Final simulation time	s
t_{max}	Maximum simulation time	s
v_{max}	Maximum initial speed	$\frac{\text{m}}{\text{s}}$
v_{x}	X-velocity of the particle	$\frac{\text{m}}{\text{s}}$
v_{x0}	Initial x-velocity	$\frac{\text{m}}{\text{s}}$
v_{y}	Y-velocity of the particle	$\frac{\text{m}}{\text{s}}$
v_{y0}	Initial y-velocity	$\frac{\text{m}}{\text{s}}$
$\mathbf{v}(t)$	Velocity	$\frac{\text{m}}{\text{s}}$
x_0	Initial x-position	m
y_0	Initial y-position	m
κ	Charge-to-mass ratio	$\frac{\text{C}}{\text{kg}}$

1.3 Abbreviations and Acronyms

Table 3: Abbreviations and Acronyms

Abbreviation	**Full Form**
A	Assumption
DD	Data Definition
GD	General Definition
GS	Goal Statement
IM	Instance Model
PS	Physical System Description

Continued on next page

Table 3: Abbreviations and Acronyms (Continued)

Abbreviation	Full Form
R	Requirement
RefBy	Referenced by
Refname	Reference Name
SRS	Software Requirements Specification
TM	Theoretical Model
Uncert.	Typical Uncertainty

2 Introduction

The motion of a charged particle through electromagnetic fields is a fundamental phenomenon arising in particle accelerators, mass spectrometers, and plasma confinement devices. It is therefore useful to have a program to simulate Charged Particle Trajectory Simulator trajectories in piecewise-uniform fields.

The following section provides an overview of the software requirements specification (SRS) for Charged Particle Trajectory Simulator. This section explains the purpose of this document, the scope of the requirements, the characteristics of the intended reader, and the organization of the document.

2.1 Purpose of Document

The primary purpose of this document is to record the requirements of Trajecto. Goals, assumptions, theoretical models, definitions, and other model derivation information are specified, allowing the reader to fully understand and verify the purpose and scientific basis of Trajecto. With the exception of system constraints, this SRS will remain abstract, describing what problem is being solved, but not how to solve it.

This document will be used as a starting point for subsequent development phases, including writing the design specification and the software verification and validation plan. The design document will show how the requirements are to be realized, including decisions on the numerical algorithms and programming environment. The verification and validation plan will show the steps that will be used to increase confidence in the software documentation and the implementation. Although the SRS fits in a series of documents that follow the so-called waterfall model, the actual development process is not constrained in any way. Even when the waterfall model is not followed, as Parnas and Clements point out [**parnasClements1986**], the most logical way to present the documentation is still to "fake" a rational design process.

2.2 Scope of Requirements

The scope of the requirements includes the analysis of two-dimensional charged particle motion in the x-y plane under the Lorentz force.

2.3 Characteristics of Intended Reader

Reviewers of this documentation should have an understanding of undergraduate level high school physics and undergraduate level calculus. The users of Trajecto can have a lower level of expertise, as explained in Sec:User Characteristics.

2.4 Organization of Document

The organization of this document follows the template for an SRS for scientific computing software proposed by [**koothoor2013**], [**smithLai2005**], [**smithEtAl2007**], and [**smithKoothoor2016**]. The presentation follows the standard pattern of presenting goals, theories, definitions, and assumptions. For readers that would like a more bottom up approach, they can start reading the instance models and trace back to find any additional information they require.

The goal statements are refined to the theoretical models and the theoretical models to the instance models.

3 General System Description

This section provides general information about the system. It identifies the interfaces between the system and its environment, describes the user characteristics, and lists the system constraints.

3.1 System Context

Fig:sysCtxDiag shows the system context. A rectangle represents the software system (Trajecto), and a circle represents the user providing inputs and receiving outputs.

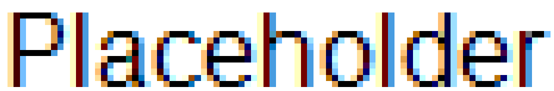

Figure 1: System Context

The user provides the particle and field parameters; the software system computes and reports the particle trajectory:

3.2 User Characteristics

The end user of Trajecto should have an understanding of undergraduate-level high school physics.

3.3 System Constraints

There are no system constraints.

4 Specific System Description

This section first presents the problem description, which gives a high-level view of the problem to be solved. This is followed by the solution characteristics specification, which presents the assumptions, theories, and definitions that are used.

4.1 Problem Description

A system is needed to predict the trajectory of a charged particle moving through piecewise-uniform electric field and magnetic field regions.

4.1.1 Terminology and Definitions

This subsection provides a list of terms that are used in the subsequent sections and their meaning, with the purpose of reducing ambiguity and making it easier to correctly understand the requirements.

- Charged particle: A particle that has a non-zero electric charge and can interact with electric and magnetic fields.
- Electric field: A physical field that exerts an electric force on charged particles.
- Magnetic field: A physical field that exerts a magnetic force on moving charged particles.
- Lorentz force: The force on a charged particle due to electric and magnetic fields.
- Trajectory: The path traced by a particle over time.
- Field region: A spatial region in which the electric field and magnetic field are specified, typically treated as uniform within the region.
- Detector line: A line at a specified location where the particle impact position is recorded.
- Velocity selector: A configuration of crossed electric and magnetic fields that allows only particles with a particular velocity to pass through without deflection.
- Cartesian coordinate system: A coordinate system that specifies each point uniquely in a plane by a set of numerical coordinates, which are the signed distances to the point from two fixed perpendicular oriented lines, measured in the same unit of length.

4.1.2 Physical System Description

The physical system of Trajecto, as shown in Fig:trajectoPhysSys, includes the following elements:

PS1: A charged particle (the simulated particle)

PS2: The electric field in the particle's current region

PS3: The magnetic field perpendicular to the x-y plane

PS4: The detector line at a specified x-coordinate

Figure 2: The physical system description

4.1.3 Goal Statements

Given the particle properties (q and m), initial conditions (x_0, y_0, $v_{\mathrm{x}0}$, $v_{\mathrm{y}0}$), the specification of electric and magnetic fields (E_{x}, E_{y}, B), and detector geometry including t_{final}, the goal statements are:

predictTrajectory: Predict the trajectory of a charged particle under prescribed electric and magnetic fields.

determineDetectorOutcome: Determine impact position and time-of-flight at a specified detector line.

4.2 Solution Characteristics Specification

The instance models that govern Trajecto are presented in the Instance Model Section. The information to understand the meaning of the instance models and their derivation is also presented, so that the instance models can be verified.

4.2.1 Assumptions

This section simplifies the original problem and helps in developing the theoretical models by filling in the missing information for the physical system. The assumptions refine the scope by providing more detail.

piecewiseUniform: The electric and magnetic fields are uniform within each field region and may change only at region boundaries.

singleParticle: The particle is treated as a point mass and point charge. (RefBy: TM:lorentzForce and TM:eqnMotion.)

noInteractions: Collisions and particle-particle interactions (including space-charge effects) are neglected.

prescribedFields: The electric and magnetic fields are user-specified and remain fixed during the simulation. (RefBy: TM:lorentzForce, TM:eqnMotion, and GD:xAccelEM.)

twoDMotion: The particle motion is confined to the x-y plane. (RefBy: GD:xAccelEM and LC:lcExtendTo3D.)

bPerpPlane: The magnetic field is perpendicular to the x-y plane. (RefBy: GD:xAccelEM.)

eAxisAligned: The electric field lies in the x-y plane and is aligned with a coordinate axis. (RefBy: GD:xAccelEM.)

rectRegions: Field-region boundaries are rectangular and parallel to the x and y axes.

lineDetector: The detector is modeled as a line located within the field region.

fullDetection: The detector line is sufficiently long to record the impact point for any trajectory within the scope of the simulation.

lorentzOnly: The particle dynamics are governed only by the Lorentz force; all other forces are neglected. (RefBy: TM:eqnMotion and UC:ucLorentzForce.)

4.2.2 Theoretical Models

This section focuses on the general equations and laws that Trajecto is based on.

Refname	**TM:velocity**
Label	Velocity
Equation	$$\mathbf{v}(t) = \frac{d\mathbf{p}(t)}{dt}$$
Description	$\mathbf{v}(t)$ is the velocity ($\frac{\text{m}}{\text{s}}$) t is the time (s) $\mathbf{p}(t)$ is the position (m)
Source	[**velocityWiki**]
RefBy	

Refname	**TM:acceleration**
Label	Acceleration
Equation	$$\mathbf{a}(t) = \frac{d\mathbf{v}(t)}{dt}$$
Description	$\mathbf{a}(t)$ is the acceleration ($\frac{\text{m}}{\text{s}^2}$) t is the time (s) $\mathbf{v}(t)$ is the velocity ($\frac{\text{m}}{\text{s}}$)
Source	[**accelerationWiki**]
RefBy	

Refname	**TM:lorentzForce**
Label	Lorentz force law
Equation	$$\mathbf{F} = q\left(E_{\text{x}} + v_{\text{y}}\right) B$$
Description	$\mathbf{F}$ is the force (N) q is the particle charge (C) E_{x} is the x-component of the electric field $\left(\frac{\text{N}}{\text{C}}\right)$ v_{y} is the y-velocity of the particle $\left(\frac{\text{m}}{\text{s}}\right)$ B is the out-of-plane magnetic flux density (T)
Notes	The electromagnetic force on a charged particle is the sum of the electric force (q*E) and the magnetic force (q*(v x B)). For 2D planar motion with an out-of-plane magnetic field B, the x-component of the Lorentz force is Fx = q*(Ex + vy*B). Assumes A:singleParticle, A:prescribedFields.
Source	–
RefBy	TM:eqnMotion

Refname	TM:eqnMotion
Label	Equation of motion under electromagnetic fields
Equation	$$a_{\text{x}} = \frac{q}{m} \left(E_{\text{x}} + v_{\text{y}}\right) B$$ $$a_{\text{y}} = \frac{q}{m} \left(E_{\text{y}} - v_{\text{x}}\right) B$$
Description	a_{x} is the x-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) q is the particle charge (C) m is the particle mass (kg) E_{x} is the x-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{y} is the y-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T) a_{y} is the y-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) q is the particle charge (C) m is the particle mass (kg) E_{y} is the y-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{x} is the x-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T)
Notes	Combining Newton's second law with the Lorentz force law, (TM:lorentzForce), the equations of motion for each velocity component are obtained. The quantity q/m is the charge-to-mass ratio. Assumes A:singleParticle, A:prescribedFields, A:lorentzOnly.
Source	–
RefBy	

4.2.3 General Definitions

This section collects the laws and equations that will be used to build the instance models.

Refname	**GD:xAccelEM**
Label	X-acceleration under electromagnetic force
Units	$\frac{\text{m}}{\text{s}^2}$
Equation	$$a_{\text{x}} = \kappa \left(E_{\text{x}} + v_{\text{y}}\right) B$$
Description	a_{x} is the x-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) κ is the charge-to-mass ratio ($\frac{\text{C}}{\text{kg}}$) E_{x} is the x-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{y} is the y-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T)
Notes	Derived from the equation of motion by applying planar motion (A:twoDMotion), out-of-plane B field (A:bPerpPlane), axis-aligned E field (A:eAxisAligned), and fixed fields (A:prescribedFields). Here $\kappa = \text{q}/\text{m}$ is defined in DD:qOverm.
Source	–
RefBy	GD:yAccelEM

Refname	**GD:yAccelEM**
Label	Y-acceleration under electromagnetic force
Units	$\frac{\text{m}}{\text{s}^2}$
Equation	$$a_{\text{y}} = \kappa \left(E_{\text{y}} - v_{\text{x}}\right) B$$
Description	a_{y} is the y-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) κ is the charge-to-mass ratio ($\frac{\text{C}}{\text{kg}}$) E_{y} is the y-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{x} is the x-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T)
Notes	Derived from the equation of motion by applying the same 2D assumptions as for GD:xAccelEM.
Source	–
RefBy	

4.2.4 Data Definitions

This section collects and defines all the data needed to build the instance models.

Refname	**DD:qOverm**
Label	Charge-to-mass ratio
Symbol	κ
Units	$\frac{\text{C}}{\text{kg}}$
Equation	$$\kappa = \frac{q}{m}$$
Description	κ is the charge-to-mass ratio ($\frac{\text{C}}{\text{kg}}$) q is the particle charge (C) m is the particle mass (kg)
Notes	The charge-to-mass ratio simplifies the equations of motion by combining the particle's charge q, and mass m into a single parameter.
Source	–
RefBy	GD:xAccelEM

4.2.5 Instance Models

This section transforms the problem defined in the problem description into one which is expressed in mathematical terms. It uses concrete symbols defined in the data definitions to replace the abstract symbols in the models identified in theoretical models and general definitions.

Refname	**IM:accelX**
Label	X-component of acceleration
Input	κ, E_{x}, v_{y}, B
Output	a_{x}
Input Constraints	
Output Constraints	
Equation	$$a_{\text{x}}=\kappa\left(E_{\text{x}}+v_{\text{y}}\right)B$$
Description	a_{x} is the x-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) κ is the charge-to-mass ratio ($\frac{\text{C}}{\text{kg}}$) E_{x} is the x-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{y} is the y-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T)
Notes	This gives the acceleration of the particle in the x-direction due to the Lorentz force (see GD:xAccelEM).
Source	–
RefBy	

Refname	**IM:accelY**
Label	Y-component of acceleration
Input	κ, E_{y}, v_{x}, B
Output	a_{y}
Input Constraints	
Output Constraints	
Equation	$$a_{\text{y}} = \kappa \left(E_{\text{y}} - v_{\text{x}}\right) B$$
Description	a_{y} is the y-acceleration of the particle ($\frac{\text{m}}{\text{s}^2}$) κ is the charge-to-mass ratio ($\frac{\text{C}}{\text{kg}}$) E_{y} is the y-component of the electric field ($\frac{\text{N}}{\text{C}}$) v_{x} is the x-velocity of the particle ($\frac{\text{m}}{\text{s}}$) B is the out-of-plane magnetic flux density (T)
Notes	This gives the acceleration of the particle in the y-direction due to the Lorentz force (see GD:yAccelEM).
Source	–
RefBy	

4.2.6 Data Constraints

The Input Data Constraints Table shows the data constraints on the input variables. The column for physical constraints gives the physical limitations on the range of values that can be taken by the variable. The uncertainty column provides an estimate of the confidence with which the physical quantities can be measured. This information would be part of the input if one were performing an uncertainty quantification exercise. The constraints are conservative to give the user of the model the flexibility to experiment with unusual situations. The column of typical values is intended to provide a feel for a common scenario.

Table 4: Input Data Constraints

Var	**Physical Constraints**	**Software Constraints**	**Typical Value**	**Uncert.**
B	–	$-B_{\text{max}} \leq B \leq B_{\text{max}}$	0.01 T	10.0%

Continued on next page

Table 4: Input Data Constraints (Continued)

Var	Physical Constraints	Software Constraints	Typical Value	Uncert.
E_{x}	–	$-E_{\text{max}} \leq E_{\text{x}} \leq E_{\text{max}}$	$0\ \frac{\text{N}}{\text{C}}$	10.0%
E_{y}	–	$-E_{\text{max}} \leq E_{\text{y}} \leq E_{\text{max}}$	$1.0 \cdot 10^{3}\ \frac{\text{N}}{\text{C}}$	10.0%
m	$m > 0$	$m_{\text{min}} \leq m \leq m_{\text{max}}$	$911.0 \cdot 10^{-33}$ kg	10.0%
q	$q > 0$	$q \leq q_{\text{max}}$	$160.0 \cdot 10^{-21}$ C	10.0%
t_{final}	$t_{\text{final}} > 0$	$t_{\text{final}} \leq t_{\text{max}}$	$1.0 \cdot 10^{-6}$ s	0.0%
v_{x0}	–	$-v_{\text{max}} \leq v_{\text{x0}} \leq v_{\text{max}}$	$1.0 \cdot 10^{6}\ \frac{\text{m}}{\text{s}}$	10.0%
v_{y0}	–	$-v_{\text{max}} \leq v_{\text{y0}} \leq v_{\text{max}}$	$0\ \frac{\text{m}}{\text{s}}$	10.0%
x_0	–	–	0 m	0.0%
y_0	–	–	0 m	0.0%

4.2.7 Properties of a Correct Solution

There are no properties of a correct solution.

5 Requirements

This section provides the functional requirements, the tasks and behaviours that the software is expected to complete, and the non-functional requirements, the qualities that the software is expected to exhibit.

5.1 Functional Requirements

This section provides the functional requirements, the tasks and behaviours that the software is expected to complete.

Input-Values: Input the values from Tab:ReqInputs.

Echo-Inputs: Echo the given inputs.

Compute-Trajectory: Compute the trajectory of the charged particle using the equations of motion derived from the Lorentz force.

Report-Outputs: Report the final position, velocity, and whether the particle reaches the detector line.

Table 5: Required Inputs

Symbol	Description	Units
B	Out-of-plane magnetic flux density	T
E_{x}	X-component of the electric field	$\frac{\text{N}}{\text{C}}$
E_{y}	Y-component of the electric field	$\frac{\text{N}}{\text{C}}$
m	Particle mass	kg
q	Particle charge	C
t_{final}	Final simulation time	s
v_{x0}	Initial x-velocity	$\frac{\text{m}}{\text{s}}$
v_{y0}	Initial y-velocity	$\frac{\text{m}}{\text{s}}$
x_0	Initial x-position	m
y_0	Initial y-position	m

5.2 Non-Functional Requirements

This section provides the non-functional requirements, the qualities that the software is expected to exhibit.

Correctness: The outputs of the code have the properties of a correct solution.

Maintainability: If a likely change is made to the finished software, it will take at most 10% of the original development time, assuming the same development resources are available.

Portability: The code shall be portable to multiple environments, particularly Windows, Mac OSX, and Linux.

6 Likely Changes

This section lists the likely changes (LC) to be made to the software.

lcExtendTo3D: A:twoDMotion - extend Trajecto to support three-dimensional (3D) motion, including a z-component of position, velocity, acceleration, electric field, and magnetic field.

7 Unlikely Changes

This section lists the unlikely changes (UC) to be made to the software.

ucLorentzForce: A:lorentzOnly - the Lorentz force law is a well-established physical principle and is unlikely to change.

8 Traceability Matrices and Graphs

The purpose of the traceability matrices is to provide easy references on what has to be additionally modified if a certain component is changed. Every time a component is changed, the items in the column of that component that are marked with an "X" should be modified as well. Tab:TraceMatAvsA shows the dependencies of the assumptions on each other. Tab:TraceMatAvsAll shows the dependencies of the data definitions, theoretical models, general definitions, instance models, requirements, likely changes, and unlikely changes on the assumptions. Tab:TraceMatRefvsRef shows the dependencies of the data definitions, theoretical models, general definitions, and instance models on each other. Tab:TraceMatAllvsR shows the dependencies of the requirements and goal statements on the data definitions, theoretical models, general definitions, and instance models.

Table 6: Traceability Matrix Sho

	A:piecewiseUniform	**A:singleParticle**	**A:noInteractions**	**A:prescribedI**
A:piecewiseUniform				
A:singleParticle				
A:noInteractions				
A:prescribedFields				
A:twoDMotion				
A:bPerpPlane				
A:eAxisAligned				
A:rectRegions				
A:lineDetector				
A:fullDetection				
A:lorentzOnly				

Table 7: Traceability Matri

	A:piecewiseUniform	**A:singleParticle**	**A:noInteractions**	**A:prescri**
DD:qOverm				
TM:velocity				
TM:acceleration				
TM:lorentzForce		X		X
TM:eqnMotion		X		X

Table 7: Traceability Matrix Sho

	A:piecewiseUniform	A:singleParticle	A:noInteractions	A:prescri
GD:xAccelEM				X
GD:yAccelEM				
IM:accelX				
IM:accelY				
FR:Input-Values				
FR:Echo-Inputs				
FR:Compute-Trajectory				
FR:Report-Outputs				
NFR:Correctness				
NFR:Maintainability				
NFR:Portability				
LC:lcExtendTo3D				
UC:ucLorentzForce				

Table 8: Traceability Matrix Showing the Connections Betw

	DD:qOverm	TM:velocity	TM:acceleration	TM:lorentzForce	TM:eqnM
DD:qOverm					
TM:velocity					
TM:acceleration					
TM:lorentzForce					
TM:eqnMotion				X	
GD:xAccelEM	X				
GD:yAccelEM					
IM:accelX					
IM:accelY					

	DD:qOverm	**TM:velocity**	**TM:acceleration**	**TM:lorentzForc**
GS:predictTrajectory				
GS:determineDetectorOutcome				
FR:Input-Values				
FR:Echo-Inputs				
FR:Compute-Trajectory				
FR:Report-Outputs				
NFR:Correctness				
NFR:Maintainability				
NFR:Portability				

The purpose of the traceability graphs is also to provide easy references on what has to be additionally modified if a certain component is changed. The arrows in the graphs represent dependencies. The component at the tail of an arrow is depended on by the component at the head of that arrow. Therefore, if a component is changed, the components that it points to should also be changed. Fig:TraceGraphAvsA shows the dependencies of assumptions on each other. Fig:TraceGraphAvsAll shows the dependencies of data definitions, theoretical models, general definitions, instance models, requirements, likely changes, and unlikely changes on the assumptions. Fig:TraceGraphRefvsRef shows the dependencies of data definitions, theoretical models, general definitions, and instance models on each other. Fig:TraceGraphAllvsR shows the dependencies of requirements and goal statements on the data definitions, theoretical models, general definitions, and instance models. Fig:TraceGraphAllvsAll shows the dependencies of dependencies of assumptions, models, definitions, requirements, goals, and changes with each other.

Figure 3: TraceGraphAvsA

Figure 4: TraceGraphAvsAll

Figure 5: TraceGraphRefvsRef

Figure 6: TraceGraphAllvsR

Figure 7: TraceGraphAllvsAll

For convenience, the following graphs can be found at the links below:

- TraceGraphAvsA
- TraceGraphAvsAll
- TraceGraphRefvsRef
- TraceGraphAllvsR
- TraceGraphAllvsAll

9 Values of Auxiliary Constants

There are no auxiliary constants.

10 References